

NORMAN WELLS AP, NT, Canada

WMO: 710430

Lat: 65.280N

Lon: 126.800W

Elev: 73

StdP: 100.46

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-39.6	-45.2	0.0	-41.8	-43.5	0.1	-39.4	11.6	-16.1	10.3	-17.9	0.6	90	0.748

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	27.6	17.1	25.7	16.2	23.8	15.4	18.1	25.6	17.1	23.7	16.1	22.2	3.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	20.1	14.3	10.3	19.5	13.3	9.6	18.9	51.3	25.7	48.3	23.7	45.5	22.3	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.4	7.6	-44.0	30.9	1.7	1.4	-45.2	32.0	-46.2	32.8	-47.1	33.6	-48.4	34.6		
			-43.9	19.5	1.5	1.0	-45.0	20.2	-45.9	20.8	-46.7	21.4	-47.8	22.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.5	-25.4	-22.5	-18.3	-5.2	7.0	15.1	17.0	13.7	6.9	-3.5	-17.1	-23.2
	DBStd	16.73	8.57	7.56	7.52	7.06	5.67	4.08	3.48	4.08	4.30	4.90	6.78	7.56
	HDD10.0	5859	1098	909	877	457	127	6	1	13	112	418	812	1028
	HDD18.3	8393	1356	1143	1136	707	353	113	65	151	344	677	1062	1286
	CDD10.0	557	0	0	0	0	33	158	219	128	18	0	0	0
	CDD18.3	48	0	0	0	0	1	14	25	8	0	0	0	0
	CDH23.3	484	0	0	0	0	17	155	227	84	1	0	0	0
	CDH26.7	91	0	0	0	0	2	31	45	13	0	0	0	0
Wind	WSAvg	2.9	2.4	2.5	3.0	3.3	3.5	3.3	3.0	3.0	3.0	2.8	2.5	2.3
Precipitation	PrecAvg	332	20	17	12	15	20	41	56	57	30	30	18	20
	PrecMax	514	62	72	38	63	65	103	117	116	86	87	39	52
	PrecMin	186	0	0	1	0	1	7	12	2	6	6	3	4
	PrecStd	76	13	15	9	12	13	21	26	31	15	18	9	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.1	-1.9	4.9	15.4	25.5	29.9	30.4	28.6	21.7	10.2	0.9	-2.6
		MCWB	-3.9	-3.5	1.3	7.8	14.0	17.2	18.6	18.2	13.6	6.0	-0.4	-3.9
	2%	DB	-7.2	-6.7	0.9	11.0	22.4	27.2	27.8	25.8	18.5	6.4	-3.0	-7.2
		MCWB	-7.9	-7.7	-1.5	5.4	12.5	16.2	17.4	17.1	12.1	4.1	-3.9	-8.0
	5%	DB	-10.8	-9.7	-3.7	7.8	19.6	25.0	25.9	23.4	16.2	4.4	-5.8	-10.1
		MCWB	-11.4	-10.6	-5.5	3.4	11.1	15.1	16.7	15.8	10.8	2.7	-6.5	-10.8
	10%	DB	-14.0	-12.2	-7.1	5.3	16.8	23.0	24.0	21.0	14.0	2.7	-7.8	-13.1
		MCWB	-14.6	-12.9	-8.4	1.7	9.5	14.3	16.1	14.7	9.9	1.4	-8.5	-13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	-3.2	1.7	8.1	14.4	18.3	19.5	19.2	14.4	7.0	-0.3	-3.9
		MCDB	-2.1	-2.0	4.6	15.0	24.6	27.1	27.5	26.5	19.8	9.1	0.8	-2.7
	2%	WB	-7.8	-7.7	-1.6	5.6	12.8	17.0	18.3	17.7	12.6	4.3	-3.8	-7.9
		MCDB	-7.1	-6.8	1.1	10.6	21.6	25.4	25.7	24.7	17.2	6.2	-3.1	-7.2
	5%	WB	-11.3	-10.5	-5.5	3.5	11.4	15.8	17.5	16.3	11.3	2.8	-6.4	-10.7
		MCDB	-10.7	-9.7	-3.8	7.6	19.1	23.4	24.1	22.0	15.6	4.2	-5.8	-10.1
	10%	WB	-14.6	-12.9	-8.5	1.7	9.8	14.8	16.6	15.2	10.1	1.5	-8.5	-13.6
		MCDB	-14.0	-12.2	-7.1	5.1	16.2	21.6	22.6	20.0	13.6	2.7	-7.9	-13.1
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.6	11.1	11.4	11.2	10.9	10.5	9.5	8.6	5.5	6.5	7.0
		MCDBR	10.1	10.5	14.3	12.9	15.7	13.9	13.6	13.5	13.1	8.0	7.4	9.4
		MCWBR	9.6	9.8	11.9	8.6	8.3	6.4	6.1	6.6	7.7	6.0	7.0	8.9
	5% WB	MCDBR	10.0	10.6	13.9	12.3	15.0	12.6	12.2	12.0	11.9	7.6	7.6	9.5
		MCWBR	9.5	9.9	11.6	8.2	8.1	6.1	5.8	6.0	7.2	5.8	7.2	9.1
Clear-Sky Solar Irradiance	taub		0.157	0.234	0.258	0.319	0.352	0.363	0.386	0.361	0.303	0.256	0.167	0.159
	taud		1.948	2.218	2.251	2.186	2.344	2.389	2.361	2.435	2.545	2.409	1.954	1.951
	Ebn at Noon		390	684	832	844	852	849	813	801	789	659	341	109
	Edh at Noon		20	51	79	110	105	103	103	87	62	42	19	6
All-Sky Solar Radiation	RadAvg		0.13	0.78	2.47	4.59	5.50	5.82	4.96	3.69	2.24	0.85	0.21	0.03
	RadStd		0.01	0.06	0.13	0.17	0.40	0.44	0.31	0.33	0.19	0.07	0.01	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.55	N/A	N/A	N/A	N/A	+0.56	-184	N/A	N/A	N/A	
Regional (4 neighbors)	+0.74	+1.50	+1.39	N/A	N/A	N/A	-236	-231	N/A	N/A	

Nomenclature: See separate page